

Vinh Ly

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Summary

Staff engineer with **18 years** building backend services, distributed systems, and web-facing products. Currently at YouTube working on video processing infrastructure. Strong in **Python** for backend and data work, with solid **Node.js** and **React** experience for full-stack features. Spent most of a long Google tenure writing Python and Go against large-scale distributed systems, and has shipped production React frontends alongside NodeAPIs.

Work Experience

Staff Software Engineer | YouTube (Google) | Remote | 02/2025 – Present

- Working on the video processing pipeline that handles millions of concurrent streams daily. Migrated legacy video ingestion services to **Python** on **Kubernetes**, cutting deployment cycle time significantly for the team.
- Designed a distributed caching layer in Go for video metadata, reducing cross-region lookups and improving playback start latency at scale.

Staff Software Engineer | Whatnot | Remote | 05/2024 – 02/2025

- Led design of a real-time bidding system using **Node.js** and **React**, serving thousands of concurrent users with sub-100ms latency during live events.
- Decomposed a Ruby monolith into **Python** and Go services, each owned by a dedicated team. This let each domain ship independently and reduced the blast radius of bad deploys.

Staff Software Engineer / Manager | Google | Remote | 03/2015 – 04/2024

- Spent the first half of this tenure as an individual contributor and technical lead. Designed distributed data pipelines that processed billions of records daily across search and indexing infrastructure.
- Managed a team of 8 engineers and continued carrying technical work. Led cross-team design reviews and drove quarterly roadmap planning with product management.
- Built internal tooling in **Python** and Go that the team relied on daily for service health monitoring and deployment orchestration.

Software Engineer | Google | Hybrid | 02/2011 – 02/2015

- Developed internal web tools for the search infrastructure team using **Python** and vanilla JavaScript, giving on-call engineers a real-time dashboard for pipeline status instead of chasing Slack messages.
- Shipped a configuration management UI in **React** that replaced a hand-edited YAML workflow, cutting the time to update service configs from about 30 minutes to under 2.

Software Design Engineer II | Microsoft | Remote | 10/2007 – 12/2011

- Built web applications for the Exchange team in C# and ASP.NET, including an internal portal used by support engineers to track ticket workflows.
- Shipped a reporting dashboard for the Exchange Online service health team. Stakeholders used it to generate weekly status digests for senior leadership.

Software Design Engineer Intern | Microsoft | Hybrid | 06/2006 – 08/2006

- Interned on the Visual Studio team. Built a code visualization feature as a demo prototype, end-to-end in C# and WPF.

Education

BS Computer Science | University of Washington | 2003 – 2007

CoreSkills

Languages: Python, Go, C++, Java, TypeScript, JavaScript, C#

WebStack: React, Node.js, HTML/CSS, REST APIs, GraphQL

Data: PostgreSQL, MySQL, Redis, MongoDB, memcached

Infra: Kubernetes, Docker, Terraform, CI/CD pipelines, GitHub Actions

Cloud: GCP (YouTube), AWS (Whatnot), Azure